Syllabus Year: 2024-2025

CHRIST (Deemed to be University), Bengaluru - 560029

End Semester Examinations - November 2025

Bachelor of Computer Applications SEMESTER - I

Course Code: BCA103-1 **Max Marks: 50**

Course Name: PRINCIPLES OF SOFTWARE DEVELOPMENT - I **Duration: 02:00**

General Instructions:

- All rough work should be done in the answer script. Do not write or scribble in the question paper except your register number.
- Verify the Course code / course title and number of pages of questions in the question paper.
- Make sure your mobile phone is switched off and placed at the designated place.
- Malpractices will be viewed very seriously.
- Answers should be written on both sides of the paper in the answer book. No sheets should be detached from the answer book.
- Answers without the question numbers clearly indicated will not be valued. No page should be left blank in the middle of the answer book.

Course Outcome: Students will be able to:

CO1: Design and Develop ethical software product.

CO2: Apply terms, concepts, and tools of relational database management systems.

CO3: Construct database projects using DDL and DML queries.

Section A **2*5=10**

1)	Analyze and differentiate between the various layers of software engineering in terms of their importance for developing ethical software products.	**2**	(CO1, L4)
2)	Analyze and differentiate between the user interface layer, application layer, and data management layer in software development.	**2**	(CO1, L4)
3)	Examine and compare the quality metrics used during the requirement elicitation phase with those applied during the software development process. Provide one example for each metric category.	**2**	(CO2, L4)
4)	Analyze and differentiate between entity and relationship in a relational database model using examples.	**2**	(CO3, L4)
5)	Analyze and differentiate between the physical and logical layers of database management systems.	**2**	(CO2, L4)

Section B **5*4=20**

6)	Construct an ethical framework for requirement elicitation that ensures quality and transparency in software development. Provide an example scenario to illustrate how this framework could be applied.	**5**	(CO1, L3)
7)	Explain the concept of a view in database management systems and illustrate its creation using DDL syntax. Further, demonstrate how data can be inserted into, updated from, and deleted from this view through appropriate DML queries.	**5**	(CO3, L2)
8) a)	Analyze and compare the use of entity and various types of views in the context of designing a library management system. Provide examples to illustrate your points.	**5**	(CO2, L4)

OR

b)	Analyze and differentiate between the data management layer and the application layer in a relational database system, providing an example to illustrate your explanation.	**5**	(CO2, L4)
9)	Construct an ethical framework for developing a software product that ensures data privacy and security at every stage of its lifecycle, including requirement gathering, design, implementation, testing, deployment, and maintenance.	**5**	(CO1, L3)

Section C **10*2=20**

10)	Construct a relational model for a Library Management System, considering entities such as Books, Authors, Publishers, Members, Borrowing, and Reservations. (i) Define each entity with its attributes. (ii) Identify the relationships between these entities and specify their cardinalities. (iii) Write SQL statements to create tables corresponding to your defined entities. (iv) Suppose a member wants to borrow a book; construct an SQL query to reserve a book for a particular member. Demonstrate how you would ensure data integrity in this process by using constraints such as foreign keys.	**10**	(CO1, L3)
11) a)	Discuss and explain the importance of requirement quality in software development projects. In your answer, describe the common sources of poor requirements and how they can impact project outcomes. Additionally, illustrate with an example a scenario where potential quality issues during requirement elicitation led to significant problems downstream in the development process.	**10**	(CO2, L2)
	OR		
b)	Discuss the concept of entity in the context of relational database management systems (RDBMS). Explain how entities are represented in the relational model, provide examples to illustrate entity types and entity sets. Further, describe the role of various views in RDBMS and explain how they can be used to present different perspectives on the same data set.	**10**	(CO2, L2)

Revised Bloom's Taxonomy (RBT) Levels

L1 - Remember
L2 - Understand
L3 - Apply
L4 - Analyse
L5 - Evaluate
L6 - Create

AI-assisted draft - teacher review and approval required before use